

KwaaiNet WebUI Manager

Complete Design Document

Version: 2.0 Date: January 2025
Philosophy: "Do Well by Doing Good"

Executive Summary

Project Vision

KwaaiNet WebUI Manager is a lightweight, standalone application serving as the primary entry point for users to discover, install, manage, and monitor KwaaiNet nodes. The platform combines distributed AI compute and storage services, enabling users to earn cryptocurrency while contributing to environmental sustainability and AI democratization.

Core Value Propositions

Do Well by Doing Good ☀️		
Join 2,847+ people who discovered you don't have to choose between making money and making a difference.		
💚 THE GOOD YOU'LL DO:		
🌱	Save 67kg CO ₂ monthly (compute + storage efficiency)	
💧	Conserve 412L water monthly vs. datacenters	
🤖	Democratize AI access AND data storage worldwide	
🌐	Build sustainable distributed infrastructure	
💰 THE WEALTH YOU'LL BUILD:		
💻	Compute: \$143/month average (RTX 3080)	
💾	Storage: \$53/month average (2TB NVMe)	
📈	Total potential: \$196/month from unused resources	
💎	Hold appreciating K tokens from dual revenue streams	
[🚀 Start Dual Earning]	[📖 Learn More]	[📊 See Economy]

Target Audience Evolution

Phase 1 - Early Adopters: Altruistic tech enthusiasts motivated by environmental impact + early economic opportunity

Phase 2 - Mainstream Users: Economics-driven adoption with proven dual-revenue ROI and simplified onboarding

TBD

☐ Map to existing persona list

Architecture Overview

Standalone Application Design

KwaaiNet-Manager/

— frontend/	# React + TypeScript WebUI
— backend/	# Lightweight FastAPI server
— installer/	# Node installation logic
— discovery/	# Multi-node discovery service
— wallet/	# K token wallet integration
— storage/	# Distributed storage fabric management
— tutorials/	# Getting started guides

Technology Stack

- Frontend: React 18 + TypeScript + Vite + Tailwind CSS
- Backend: FastAPI (minimal, separate from nodes)
- Distribution: Electron app or Tauri for cross-platform
- Charts: Chart.js or Recharts for visualizations
- WebSockets: Real-time updates and live metrics
- Database: Embedded SQLite for configuration storage

Key Architectural Principles

- Separation of Concerns: WebUI manager independent of kwaainet-node processes
- Lightweight: Under 50MB total installation footprint
- Multi-Service Support: Discover and manage compute + storage services
- Real-Time Updates: Live metrics, earnings, and network status
- Offline Capability: Core functions work without internet connectivity

Progressive Authentication & Hardware Testing

Zero-Commitment Entry Experience

Welcome to KwaaiNet! 🙌

🌐 See what your computer could earn in our network

No signup required • No commitment • Takes 3 minutes

[🔍 Test My Computer] [👁️ Just Browse First]

💡 2,847 people worldwide are already earning while making AI more sustainable. See what's possible...

Hardware Benchmarking Flow

Step 1: Permission-Based Hardware Discovery

Let's Test Your Setup! 💻

We'll benchmark your hardware to show realistic earnings

🔒 Your Privacy:

- All tests run locally on your machine
- No personal data collected
- No software installed
- You can stop anytime

📊 What we'll test:

- ✓ GPU compute power (if available. You don't need one to join.)
- ✓ CPU performance
- ✓ Storage speed & capacity
- ✓ Network speed & latency

🎯 The community average: 50 tokens/sec + 2TB storage
Let's see how yours compares!

[🚀 Start Benchmark] [? Learn More] [⬅️ Back]

Step 2: Real-Time Dual Benchmarking

Complete Benchmark Running... ⚡💾

🔥 Testing your RTX 3080 + 2TB NVMe...

└ Your Performance

 GPU: 67.3 tok/s

87%

└ Community Average

 RTX 3080: 65.2 tok/s

85%

 Storage: 3.2 GB/s

94%

 NVMe Avg: 2.8 GB/s

82%

🏆 Grade: A+ (Both!)

Power: 285W + 8W 50 tok/s + 1TB

 Network Average:

Progress: [██████████] Complete

 Projected Dual Earnings: \$196/month

 Environmental Impact: -67kg CO₂, -412L H₂O

[ See Full Results] [ Pause Test]

Step 3: Comprehensive Results Dashboard

Your Complete KwaaiNet Profile

 **Benchmark Complete! Dual earning potential:**

Compute Profile

GPU: RTX 3080 (67.3 tok/s)
CPU: i7-12700K (8.7 tok/s)
Network: 45ms (Excellent)
Performance: A- Grade
Rank: #847 of 2,847

Storage Profile

Drive: 2TB NVMe (A+ tier)
Available: 1.8TB usable
Speed: 3.2GB/s read
Health: 96/100 score
Rank: #234 of 1,923

 **Combined Monthly Earning Potential:**

	Conservative	Realistic	Optimized
Compute:	\$118	\$143	\$167
Storage:	\$41	\$53	\$61
Total:	\$159	\$196	\$228

 Green bonus potential: +\$29 (solar detected)

 Quality bonus potential: +\$23 (high ratings)

 Referral income: \$15-50/month (network effects)

 **Environmental Impact (Monthly):**

- CO₂ Savings: 67kg (compute + storage efficiency)
- Water Conservation: 412L vs traditional datacenters
- Energy Efficiency: +31% vs centralized infrastructure

 Start Earning Both

 Optimize Setup

 Compare

Authentication Trigger Points

Trigger 1: Save Benchmark Results

Save Your Complete Profile?  

 Impressive results! Your setup could earn \$196/month from BOTH compute (\$143) AND storage (\$53)!

 Create a free account to:

- ✓ Save your dual-performance profile
- ✓ Get notified of high-paying storage jobs
- ✓ Track compute + storage earnings together
- ✓ Access premium storage marketplace
- ✓ Monitor both performance metrics
- ✓ Receive optimization recommendations

 Email: (use passkeys if available)

 Create password: (validate email)

 Node name: (generate a random username)

 Account Benefits:

- Priority access to high-value storage contracts
- Advanced analytics for both revenue streams
- Dual-earning optimization recommendations

☒ Create Account

☐  Google Sign-In

☐  Skip

 Your storage data stays encrypted and private

Trigger 2: Node Installation

Ready to Earn from Compute + Storage?  

You're about to become a dual-service provider:

-  COMPUTE NODE: Process AI workloads
-  STORAGE NODE: Provide distributed storage fabric

Account needed for:

-  Dual K token wallets (compute + storage earnings)
-  Combined performance monitoring
-  Unified reputation system
-  Integrated marketplace access
-  Encrypted storage management

Quick Setup:

-  Email: [your-email@example.com]
-  Node Name: [my-dual-earner]
-  Storage Allocation: [1.8TB] of [2TB available]
-  Secure Password: [.....] (use passkeys if possible)

 Setup Dual Node

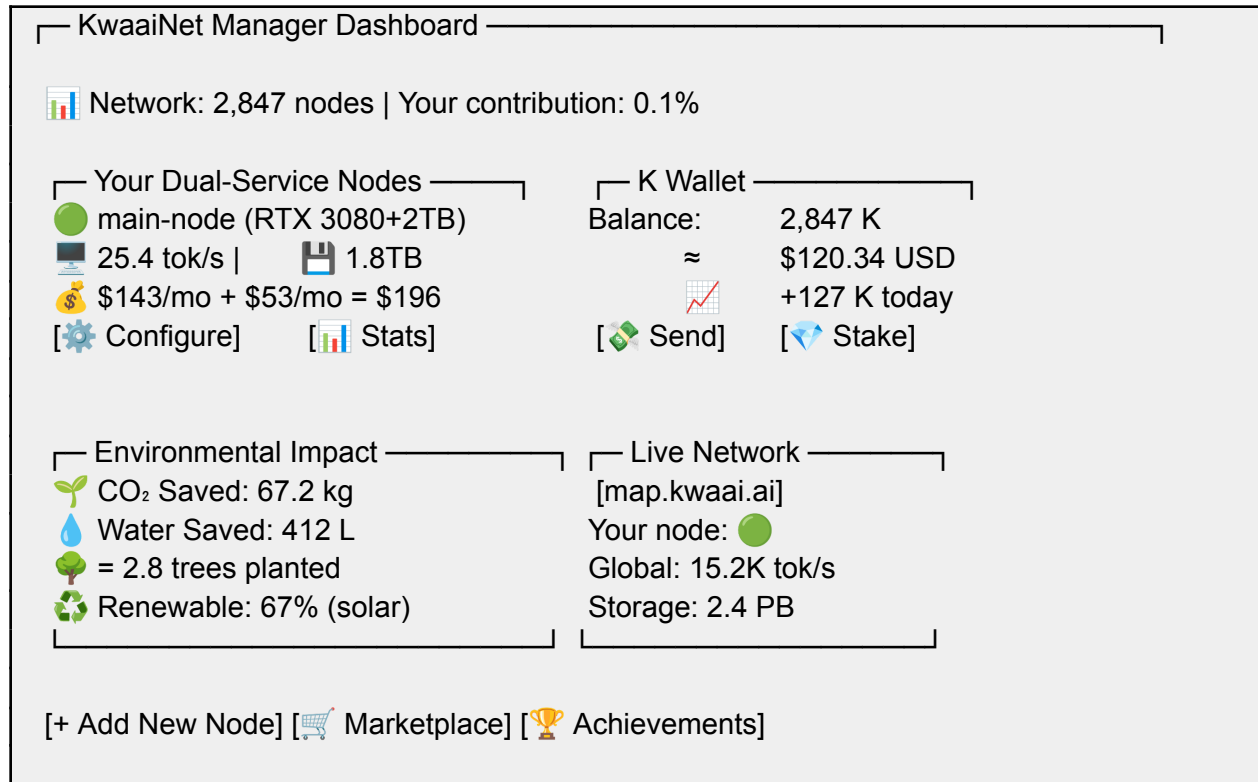
 Google Sign-In

 Start earning from TWO revenue streams immediately!

 Expected first-week earnings: \$45-65 combined

User Interface Design

Main Dashboard Layout



Dual Earnings Dashboard

Your Earning Breakdown 💰💾

Monthly Revenue Streams:

Compute Earnings

GPU Processing: 2,234 K

CPU Tasks: 156 K

Green Bonus: 347 K

Quality Bonus: 123 K

Total: 2,860 K ≈ \$120.92

Storage Earnings

File Storage: 847 K

CDN Hosting: 234 K

Model Cache: 189 K

Data Transit: 67 K

Total: 1,337 K ≈ \$56.53

Combined Monthly Total: 4,197 K ≈ \$177.45

Storage Utilization:

Allocated: 1.8TB of 2TB SSD (90% efficient)

Hot Storage: 847GB (high-value, frequent access)

Warm Storage: 734GB (regular access)

Cold Archive: 219GB (backup, infrequent access)

Combined Performance Tier: A+ (Top 15% network-wide)

Storage Details

Optimize Mix

Usage Trends

9

Dual Marketplace Economy

KwaaiNet Compute + Storage Economy

Network Stats (24h):

Compute Market

Nodes: 2,847

Rate: 15.2K tok/s

Volume: \$127K

Storage Market

Providers: 1,923

Capacity: 2.4 PB

Volume: \$89K

Current Rates:

Compute (per hour)

A100 GPU: 15.7 K

RTX 4090: 8.2 K

RTX 3080: 4.1 K

CPU-only: 0.8 K

Storage (per TB/month)

Hot NVMe: 18.9 K

Hot SSD: 15.2 K

Warm SSD: 8.4 K

Cold Arch: 2.1 K

Live Opportunities:

Llama-3.1-70B hosting: 140GB @ 18.9 K/TB

Training dataset cache: 2.3TB @ 15.2 K/TB

High-demand compute: RTX 3080+ needed now!

Your potential: \$143/mo compute + \$53/mo storage

List Storage

List Compute

Browse Jobs

Environmental Impact Integration

Personal Impact Tracking

Your Environmental Impact

 Making AI sustainable, one node at a time:
Your RTX 3080 + 2TB Storage vs Traditional Datacenter:

	You	Datacenter	Savings
Energy/1M tokens	2.8 kWh	3.7 kWh	-24% 
CO ₂ /1M tokens	1.2 kg	1.9 kg	-37% 
Water/1M tokens	0.8 L	2.1 L	-62% 
Storage efficiency	+31%	baseline	+31% 

 Your Monthly Environmental Impact:

- CO₂ Avoided: 67kg = 2.8 trees planted
- Water Saved: 412L = 27 dishwasher cycles
- Energy Efficiency: +31% vs datacenter

 Annual Impact:

- 804kg CO₂ saved = 36 trees planted
- 4,944L water conserved = 329 dishwasher loads
- Support renewable energy adoption

 Community Impact (Your contribution):

- Network total: 1,847 tons CO₂ saved/year
- Your share: +0.044% of global savings
- Help maintain position as #2848 greenest nodes

 Green Achievements:

-  Eco Starter - First green inference
-  Water Guardian - Saved first 1,000L
-  Tree Planter - CO₂ savings = 1 tree planted
-  Climate Champion - Save 1 ton CO₂ (Progress: 8%)

 Environmental Leaders Impact Trends Share

Green Achievement System

EcoAI Achievements 🏆🌱

☀️ Your Green AI Journey:

- ✅🌱 Eco Starter - First green inference
- ✅💧 Water Saver - Saved first 100L water
- ✅🌳 Tree Planter - CO₂ savings = 1 tree planted
- 🔄♻️ Renewable Runner - 50% renewable energy (67/100)
- 🔄💾 Efficient Storage - Top 25% storage efficiency
- 🔒🌍 Climate Hero - Save 1 ton CO₂ (Progress: 8%)
- 🔒💡 Sustainability Legend - Top 1% green contributors

🏆 Monthly Green Challenges:

🎯 January: "New Year, Greener AI"

- Save 500kg CO₂ collectively - Reward: 2x points
- Progress: ██████████░░░░ 87% (1,734 participants)

🌊 February: "Water Warriors"

- Conserve 50,000L as a community
- Unlock: Advanced water tracking features

📊 Your Green Stats vs. Community:

- CO₂ Saved: 67.2kg (Better than 73% of users) ☀️
- Water Saved: 412L (Better than 81% of users) ☀️☀️
- Renewable %: 67% (Top 15%!) ☀️☀️☀️
- Green Streak: 12 days (Top 10%!) ★

[🏆 View All Badges](#) [🌍 Join Challenge](#) [📢 Share Win](#)

Development Implementation

Phase 1: Foundation (Weeks 1-4)

Phase 1: Foundation 🏗️

🎯 Goal: Core WebUI with dual-service management

📦 Deliverables:

- ✓ React + TypeScript project setup
- ✓ Basic dashboard with compute + storage discovery
- ✓ K token wallet integration
- ✓ Node start/stop/restart for both services
- ✓ Live map.kwaai.ai integration
- ✓ Environmental impact calculator
- ✓ Hardware benchmarking system (compute + storage)

🎯 Success Criteria:

- Can discover and control existing dual-service nodes
- Real-time metrics for compute + storage display
- K token balance and transaction display
- Environmental impact tracking active
- Guest mode hardware testing functional

📊 Target Metrics:

- 100+ beta testers
- <3s load time
- 95% hardware detection success

Duration: 4 weeks

Phase 2: Enhanced Experience (Weeks 5-8)

Phase 2: Enhanced Experience ✨

🎯 Goal: Full installation wizard and economic features

📦 Deliverables:

- ✓ Complete installation wizard (Express + Advanced)
- ✓ Full K token wallet with transaction history
- ✓ Dual marketplace integration (compute + storage)
- ✓ Real-time earnings dashboard for both services
- ✓ Environmental achievement and gamification system
- ✓ Community features and social sharing

🎯 Success Criteria:

- New users install dual nodes in <5 minutes
- Wallet shows detailed earnings by service type
- Environmental gamification drives engagement
- Community features increase retention

📊 Target Metrics:

- 1,000+ active users
- 70% 7-day retention
- 90% installation success rate

Duration: 4 weeks

Phase 3: Optimization (Weeks 9-12)

└─ Phase 3: Advanced Features 🚀

🎯 Goal: Optimization and advanced functionality

📦 Deliverables:

- ✓ Multi-node management interface
- ✓ Advanced monitoring and analytics
- ✓ DeFi integrations (staking, liquidity mining)
- ✓ Mobile-responsive design
- ✓ Automated update system
- ✓ Performance optimization tools
- ✓ Storage quality monitoring and alerts

🎯 Success Criteria:

- Power users manage multiple dual nodes efficiently
- Advanced analytics provide actionable insights
- Mobile usage >20% of sessions
- Update system >95% success rate



Target Metrics:

- 5,000+ active users
- 40% 30-day retention
- \$150+ average monthly earnings per user

Duration: 4 weeks

Phase 4: Production Ready (Weeks 13-16)



Phase 4: Production Ready



Goal: Polish, scale, and launch preparation



Deliverables:

- ✓ Comprehensive testing and bug fixes
- ✓ Performance optimization and caching
- ✓ Documentation and user guides
- ✓ A/B testing framework
- ✓ Analytics and telemetry
- ✓ Localization (top 5 languages)



Success Criteria:

- Handles 10,000+ concurrent users
- <2s load times on average hardware
- >90% user satisfaction in surveys
- Ready for public launch



Launch Targets:

- 10,000+ users in first month
- \$2M+ network transaction volume
- 50+ new nodes daily

Duration: 4 weeks

Success Metrics

Primary KPIs

Success Metrics Dashboard

Primary KPIs:

Adoption Metrics:

- Daily Active Users: Target 1,000+ by Phase 2
- Node Installations: 500+ dual-service nodes/week
- User Retention: >70% (7-day), >40% (30-day)
- Time to First Earning: <24 hours for 80% of users

Economic Metrics:

- Total K Tokens Earned: Cumulative network earnings
- Average Dual Revenue: \$150+/month per node
- Wallet Adoption: >80% integrated wallet usage
- Marketplace Volume: \$25K+/month (compute + storage)

Environmental Metrics:

- CO₂ Savings: Cumulative emissions avoided
- Water Conservation: Total usage reduction
- Green Node Adoption: >50% renewable energy
- Environmental Engagement: >30% share achievements

Technical Performance:

- Load Time: <3 seconds average
- WebSocket Uptime: >99.5% reliability
- Discovery Success: >95% auto-detection
- Installation Success: >90% first-time success

User Experience:

- Net Promoter Score: >50 target
- Support Volume: <5% users need help
- Feature Usage: Monitor dual-service adoption
- Community Engagement: Discord participation rates

 View Details] [ Trends] [ Set Targets]

Competitive Positioning

Market Differentiation

Competitive Advantages 🏆

🚀 Unique Value Propositions:

vs vs Compute-Only Networks (Render, Akash):

- ✓ Dual revenue: compute + storage earnings
- ✓ Higher utilization: complementary service demand
- ✓ More stable income: storage provides base revenue
- ✓ Lower latency: co-located compute and data

vs vs Storage-Only Networks (Filecoin, Storj):

- ✓ Additional compute revenue stream
- ✓ Smart data processing at storage location
- ✓ Higher value services beyond storage
- ✓ AI-first use cases vs generic file storage

vs vs Traditional Cloud (AWS, GCP, Azure):

- ✓ 31% more energy efficient distributed architecture
- ✓ 60-80% lower costs through P2P model
- ✓ Geographic edge distribution by default
- ✓ Community ownership vs corporate control
- ✓ Transparent environmental impact

☀️ "Do Well by Doing Good" Philosophy:

- ✓ First platform quantifying environmental impact
- ✓ Gamified sustainability creates emotional connection
- ✓ Community-driven vs purely transactional
- ✓ Values alignment drives long-term retention

🔬 Progressive Authentication Innovation:

- ✓ Industry-first "try before buy" with real projections
- ✓ Zero-commitment hardware testing builds trust
- ✓ Permission-based privacy respects user autonomy

[📊 Market Analysis] [🎯 Positioning] [📈 Roadmap]

Conclusion

— KwaaiNet: The Future of Distributed Computing 🚀 —

The KwaaiNet WebUI Manager represents a fundamental evolution in distributed computing platforms.

☀️ Key Innovations:

💰 Dual Revenue Optimization:

First platform to seamlessly integrate compute and storage earnings, maximizing ROI from existing hardware

🔬 Hardware-First Onboarding:

Realistic projections based on actual user hardware before any commitment - builds trust and sets proper expectations for sustainable growth

🌍 Environmental Gamification:

Quantified impact creates meaning beyond profit, transforming participation into values-aligned movement

👥 Community Integration:

Live network visualization with map.kwaai.ai creates belonging and demonstrates real global participation

🔒 Progressive Authentication:

Value delivered before authentication required - revolutionary approach to user acquisition

"Do Well by Doing Good" Philosophy:

Eliminates false choice between profit and purpose - users discover they can maximize both simultaneously

🎯 Market Positioning:

Captures emerging distributed AI infrastructure market while building community that values sustainability and democratized access to computing resources

Success = sustainable, profitable, environmentally beneficial network that truly democratizes AI

[🚀 Begin Implementation] [📋 Review Checklist]